Computer Programming 1

Integrated Development Environments - IDEs

IDE

- Several Integrated Development Environments (IDEs) exist to support the developers in writing the code, making it “easy” and quick
  - E.g., Eclipse, Visual Studio Code, NetBeans, SonarLint, JetBrains, ….
- An IDE provides a suite of tools and features that aid development: code editor, easy compilation setup, a debugging tool, automatic execution, etc.
- Some IDEs are specific for a give programming language, others are more generic and can be configured to work with different languages
  - In this last case, a pre-configuration step is needed to setup the environment before being able to work with the desired language, e.g., setting the compiler to be used, the text editor

Use of the IDE in the course

- In the course, we will see how to setup and use Visual Studio Code as IDE for developing with C++ language programming



Visual Studio

https://code.visualstudio.com

- The setup of the IDE depend on the environment.
  - Too many combinations are possible!
  - As example, we consider:
    - *Windows OS*
    - *WSL with g++ compiler*
    - *Visual Studio Code on Windows OS*

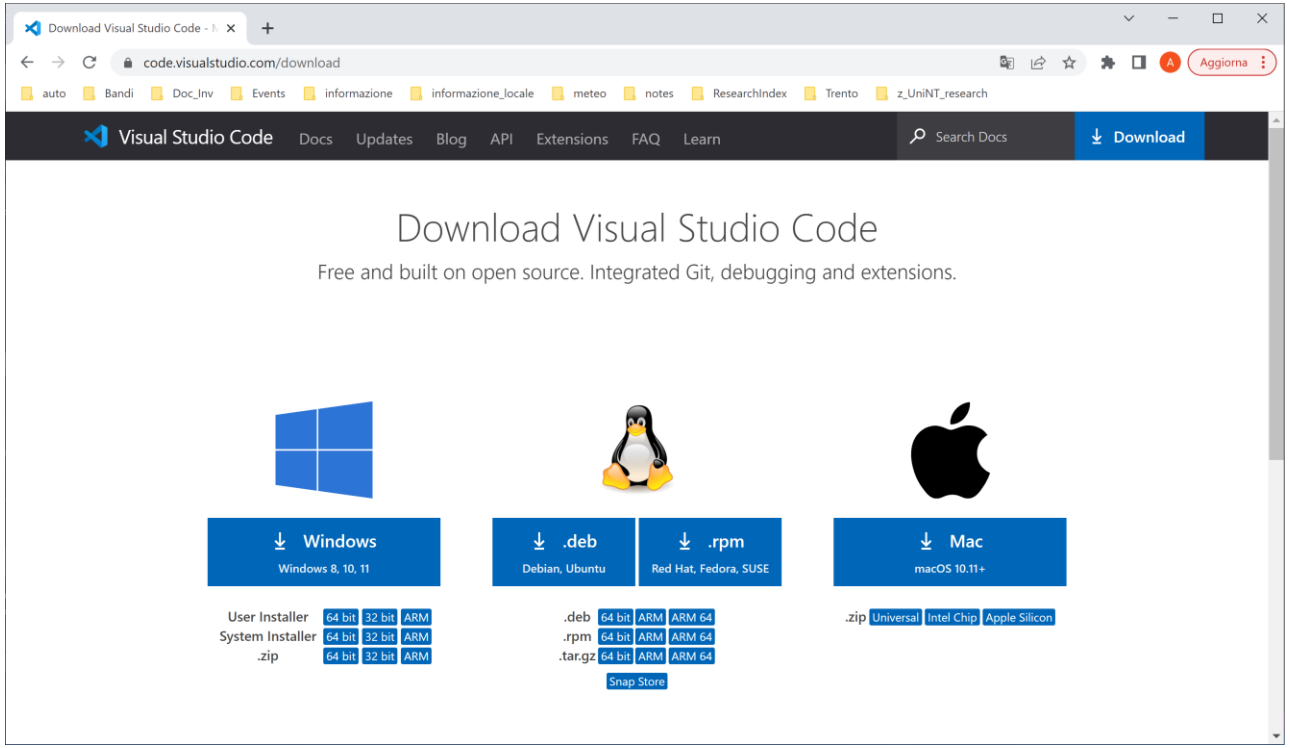
Visual Studio Code installation (1)

- Platform assumption:
  - Windows OS
  - WSL already installed
  - g++ compiler already installed in WSL

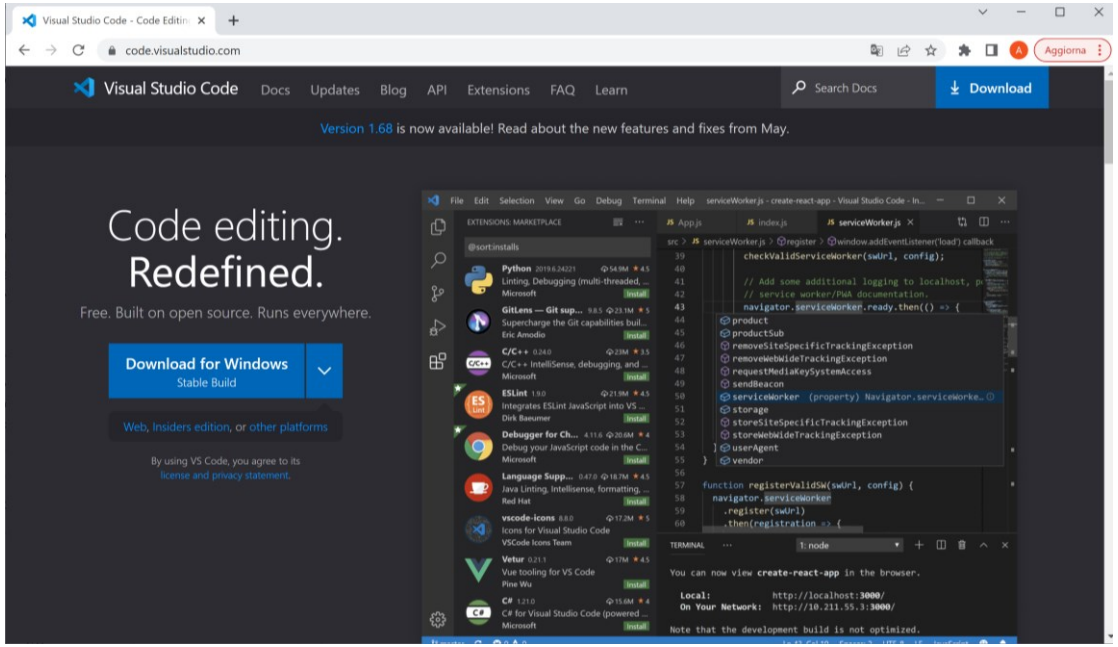
- Steps:
  1. Download and Install Visual Studio Code
  2. Conclude the installation in Visual Studio Code
     a) Install the C++ extension
     b) Install the Remote-WSL extension
  3. Test the installation
     a) Write code in Visual Studio Code
     b) Run the code with the WSL g++ compiler
     c) Add source code, compile and run it
     d) Debug code

Useful links:
https://visualstudio.microsoft.com/it/downloads/
https://code.visualstudio.com/docs/cpp/config-msvc
https://code.visualstudio.com/docs/cpp/config-wsl
https://answers.microsoft.com/en-us/insider/forum/all/how-to-enable-the-windows-subsystem-for-linux/16e8f2e8-4a6a-4325-a89a-fd28c7841775

1.Download and install Visual Studio Code (1)

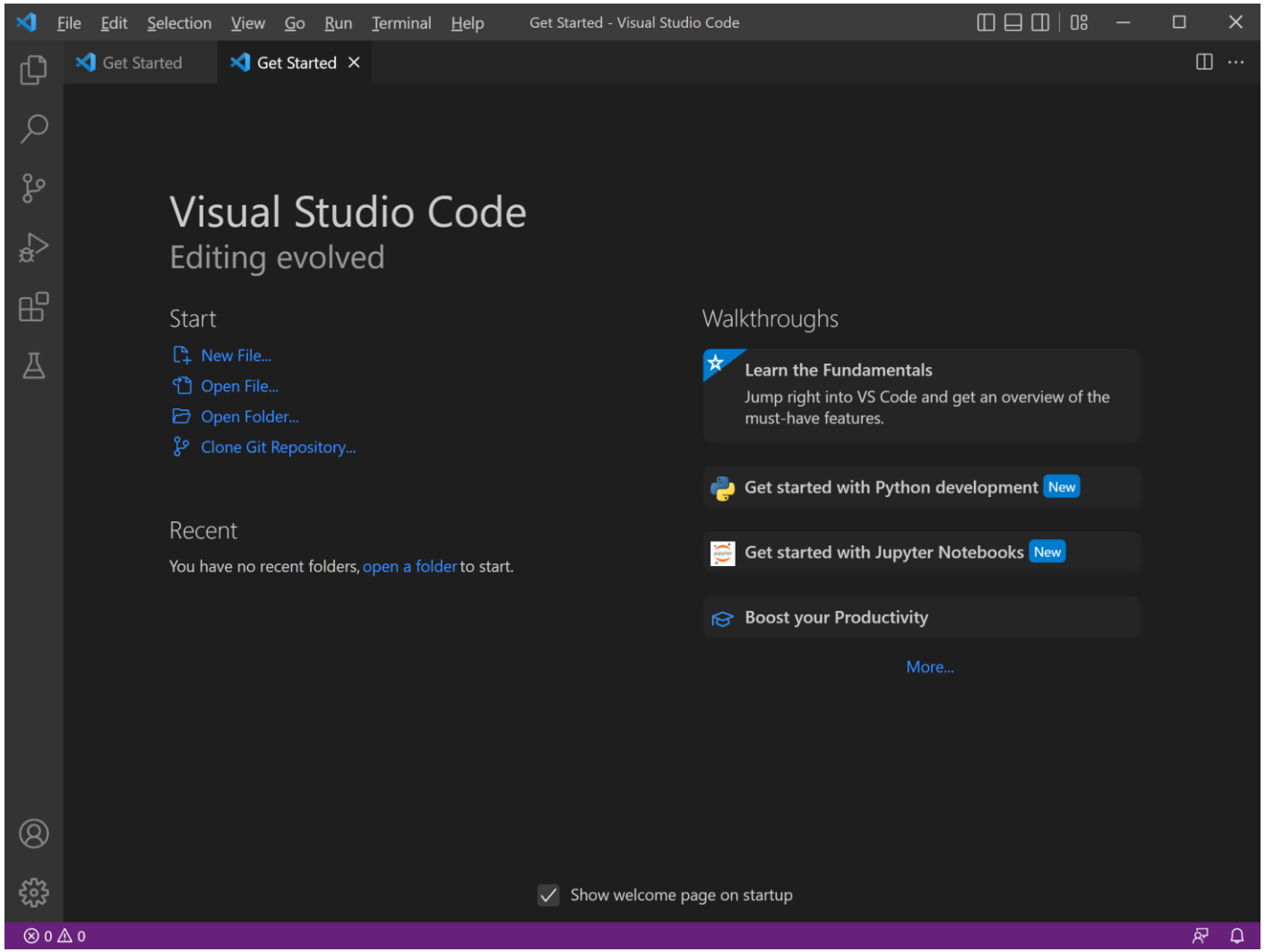


Download and install the version for your OS



https://code.visualstudio.com/docs/?dv=win

1.Download and install Visual Studio Code (2)



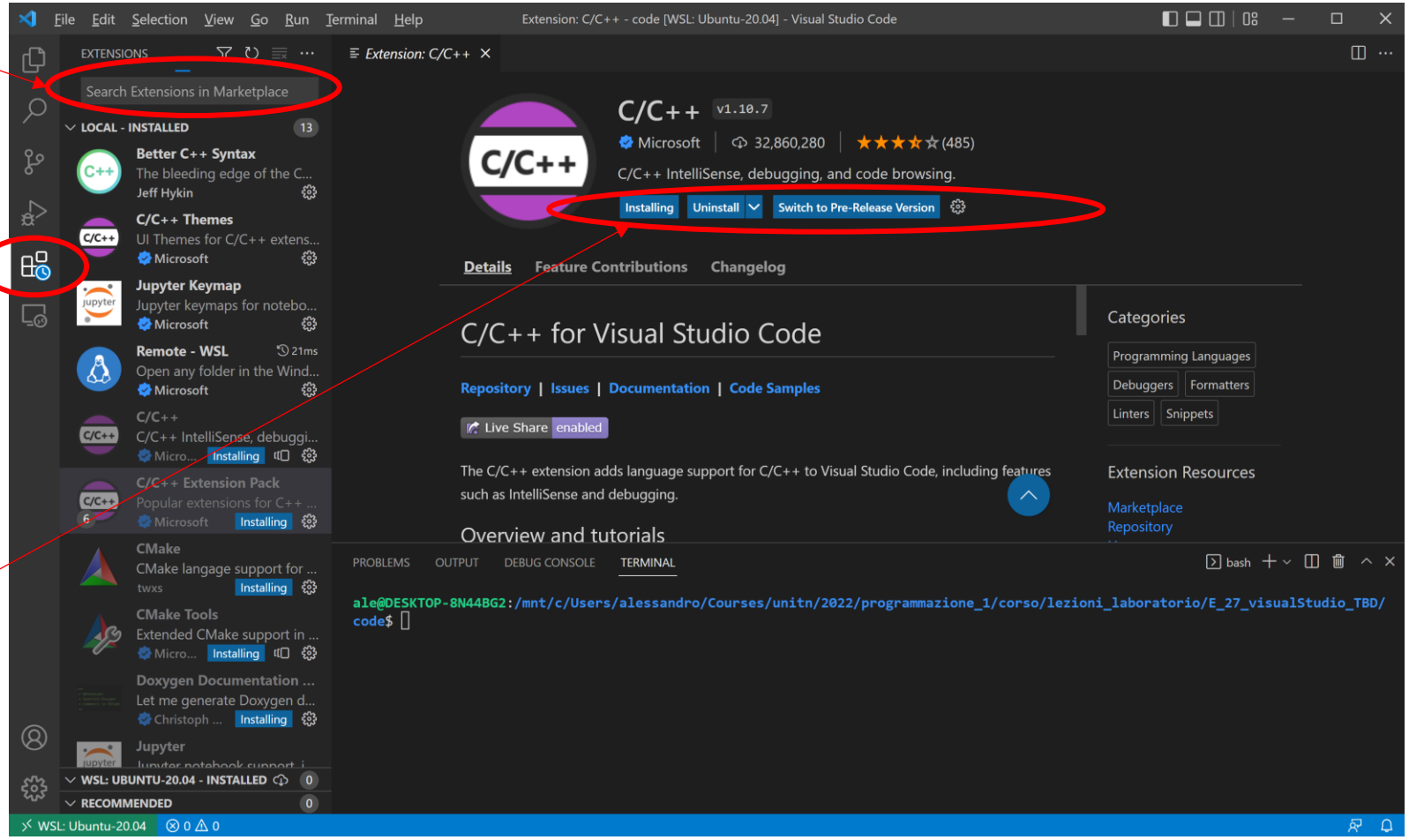
Installation concluded.
Start the Visual Studio Code environment.

2.a) In Visual Studio install the C++ extension

2) Search for C++

1) Extension

3) Install button



Install the C++ extension

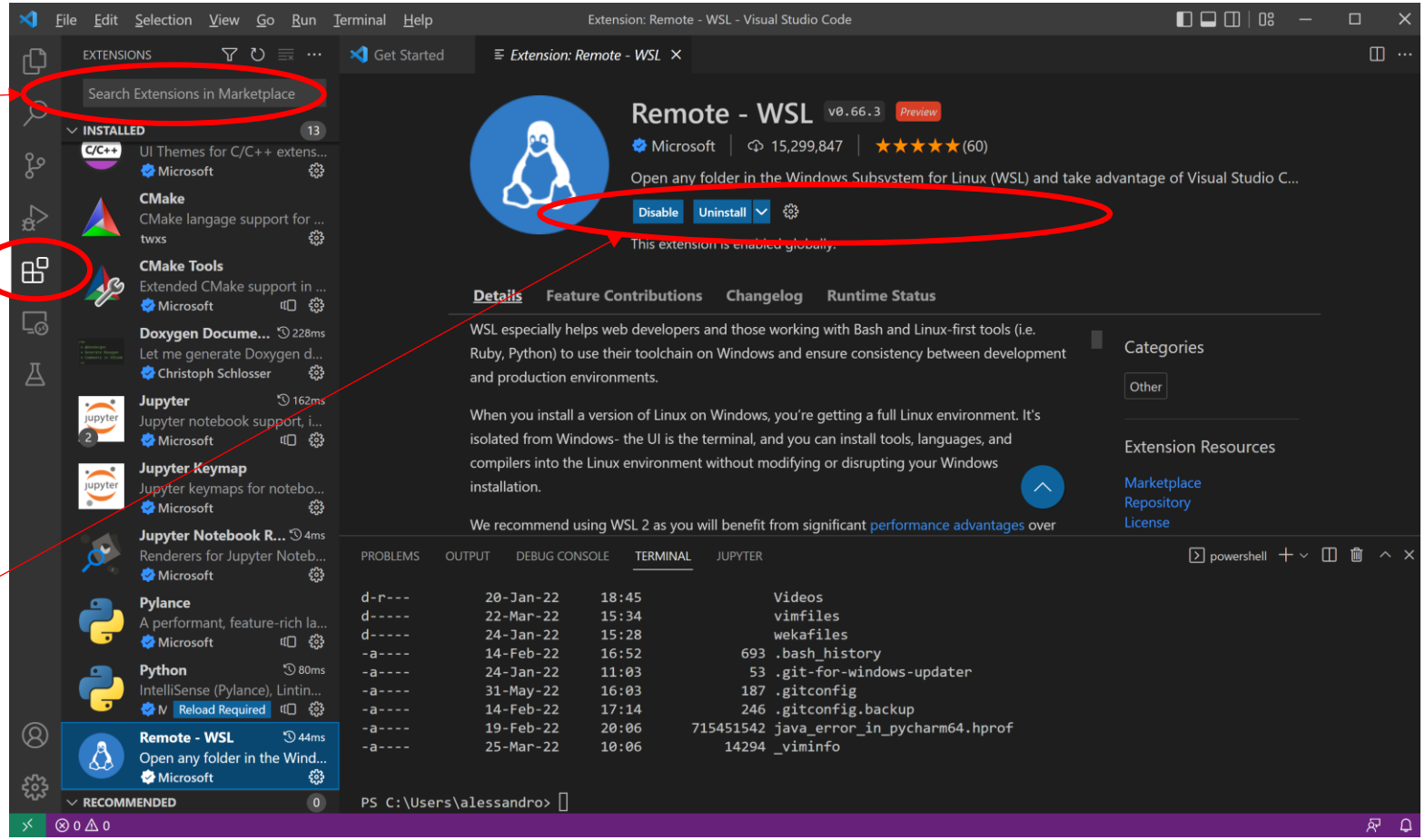
https://marketplace.visualstudio.com/items?itemName=ms-vscode.cpptools

2.b) In Visual Studio install the WSL-Remote extension

2) Search for Remote WSL

1) Extension

3) Install button

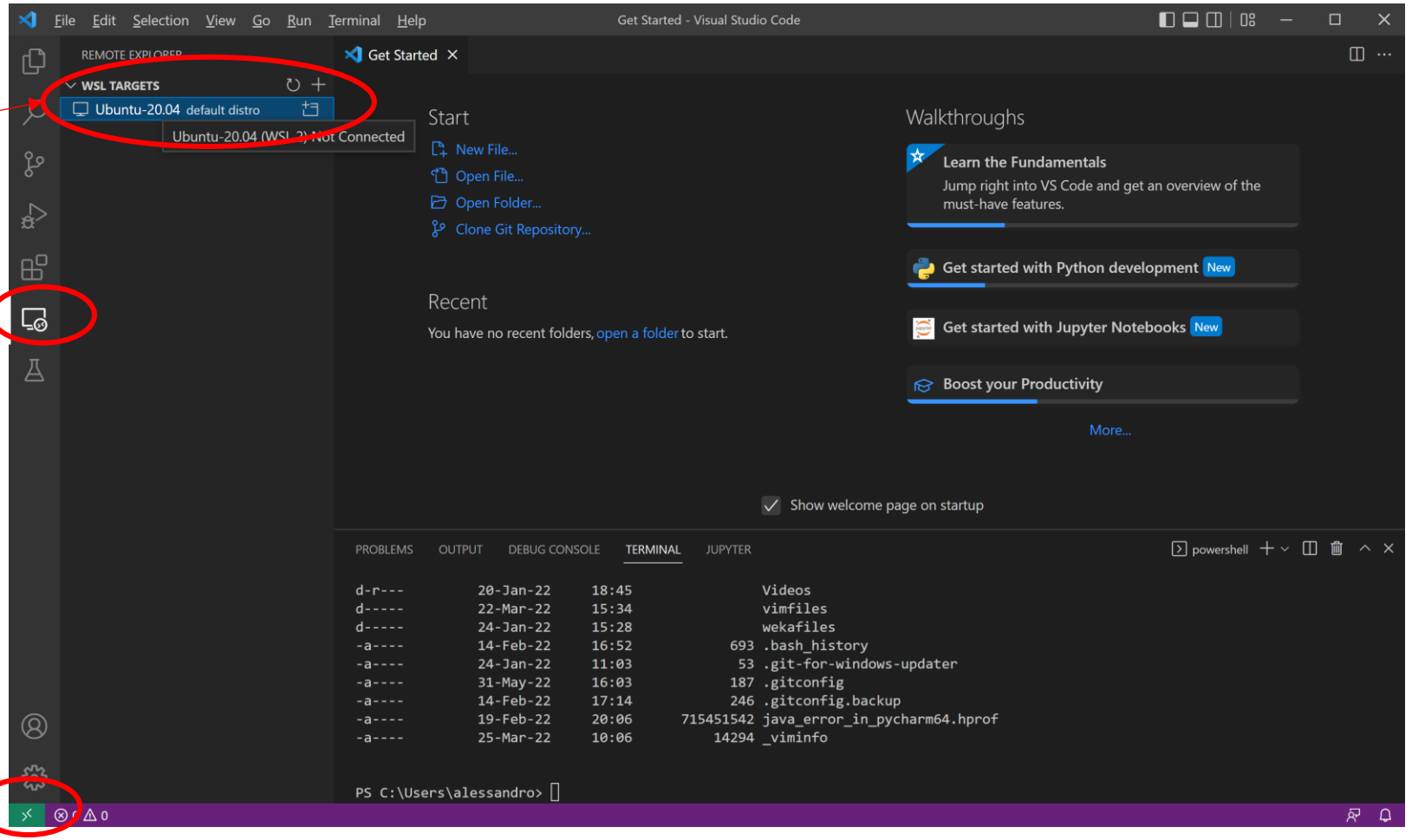


Install the Remote WSL extension
(note: WSL needs to be enabled in your OS*)

https://code.visualstudio.com/docs/cpp/config-wsl

*https://answers.microsoft.com/en-us/insider/forum/all/how-to-enable-the-windows-subsystem-for-linux/16e8f2e8-4a6a-4325-a89a-fd28c7841775

3.a) Connect to Remote WSL (1)



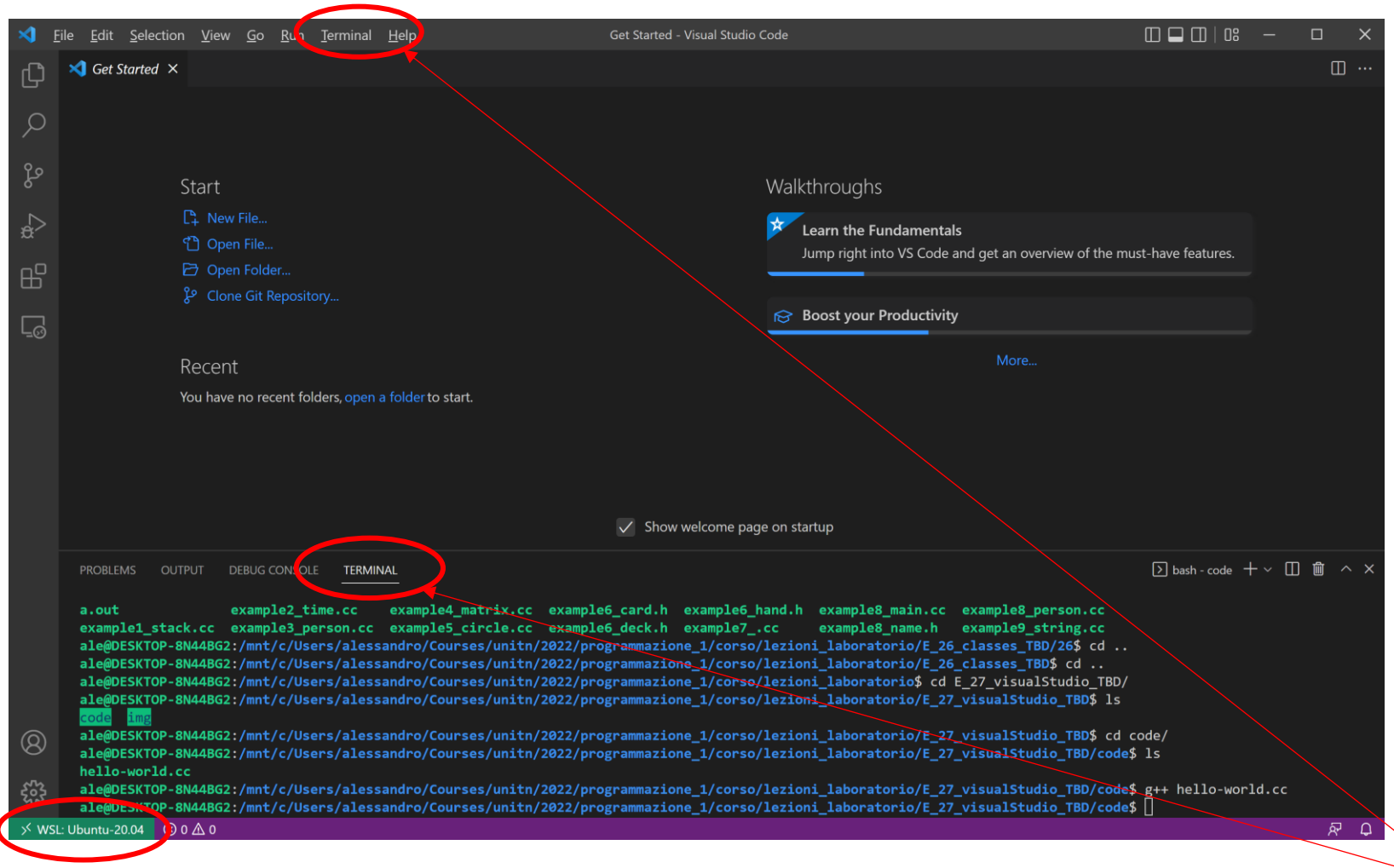
3) Start connection

1) Check the connection with the Remote WSL: now not connected
2) Click to start the connection

Start the remote WSL

(note: WSL needs to be enabled in your OS – see the previous slide)

3.b) Write the code (1)



Open the WSL terminal.
Check the WSL environment (e.g., traverse the disk and check the g++ command).

In case it is needed, install the g++ compiler in the WSL terminal:

sudo apt-get update
sudo apt install aptitude -y
sudo aptitude install g++

1) Check the connection with the Remote WSL: now connected

2) Open the WSL terminal and check if the WSL is working as expected

3.b) Write the code (2)

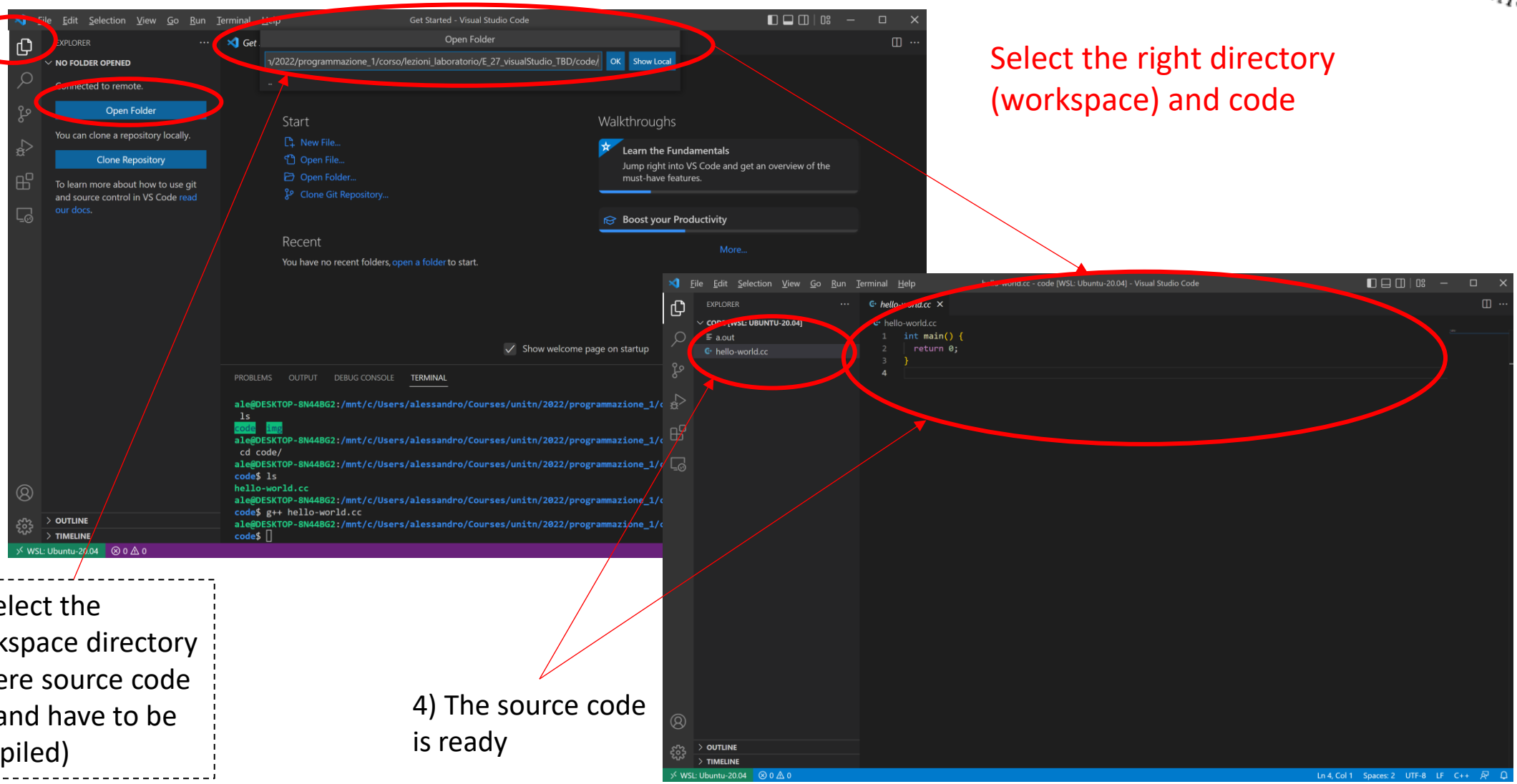
1) Selected the right working area and code, if any

2) Click Open Folder

3) Select the workspace directory (where source code are and have to be compiled)

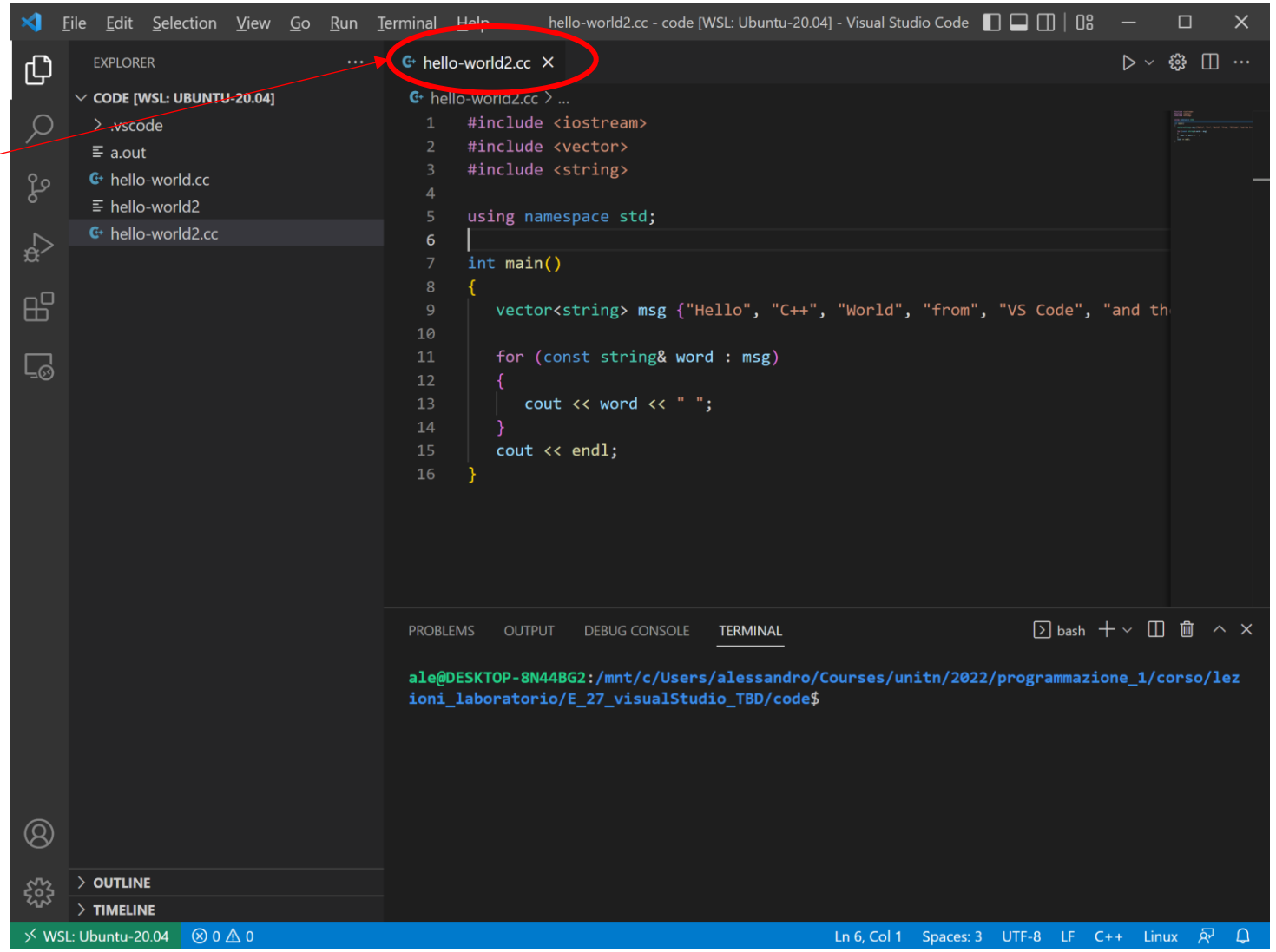
Select the right directory (workspace) and code

4) The source code is ready



3.b) Write the code (3)

Coding



Coding …

E.g.,

```
#include <iostream>

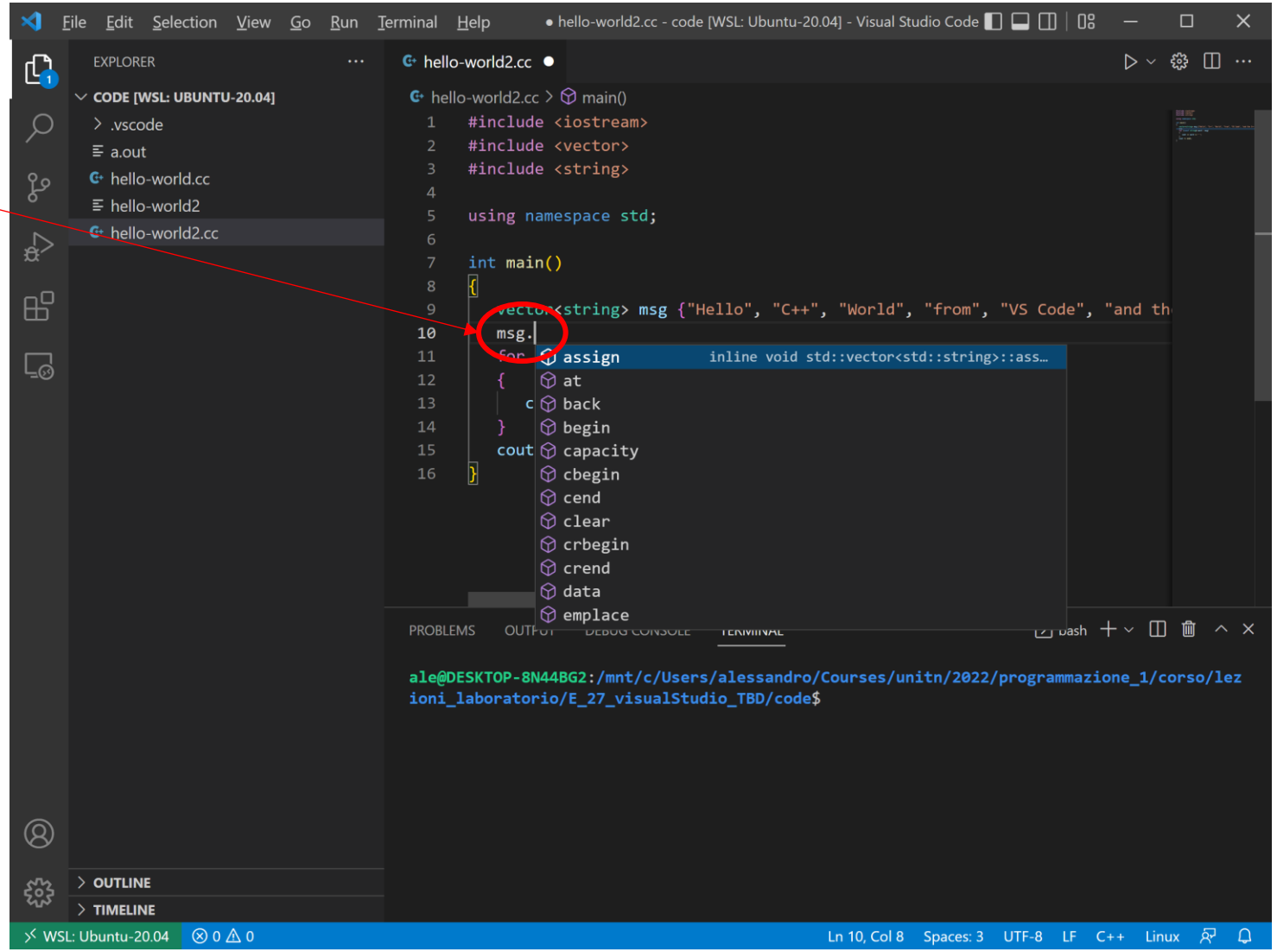
using namespace std;

int main()
{
   cout << “Hello World” << endl;

}
```

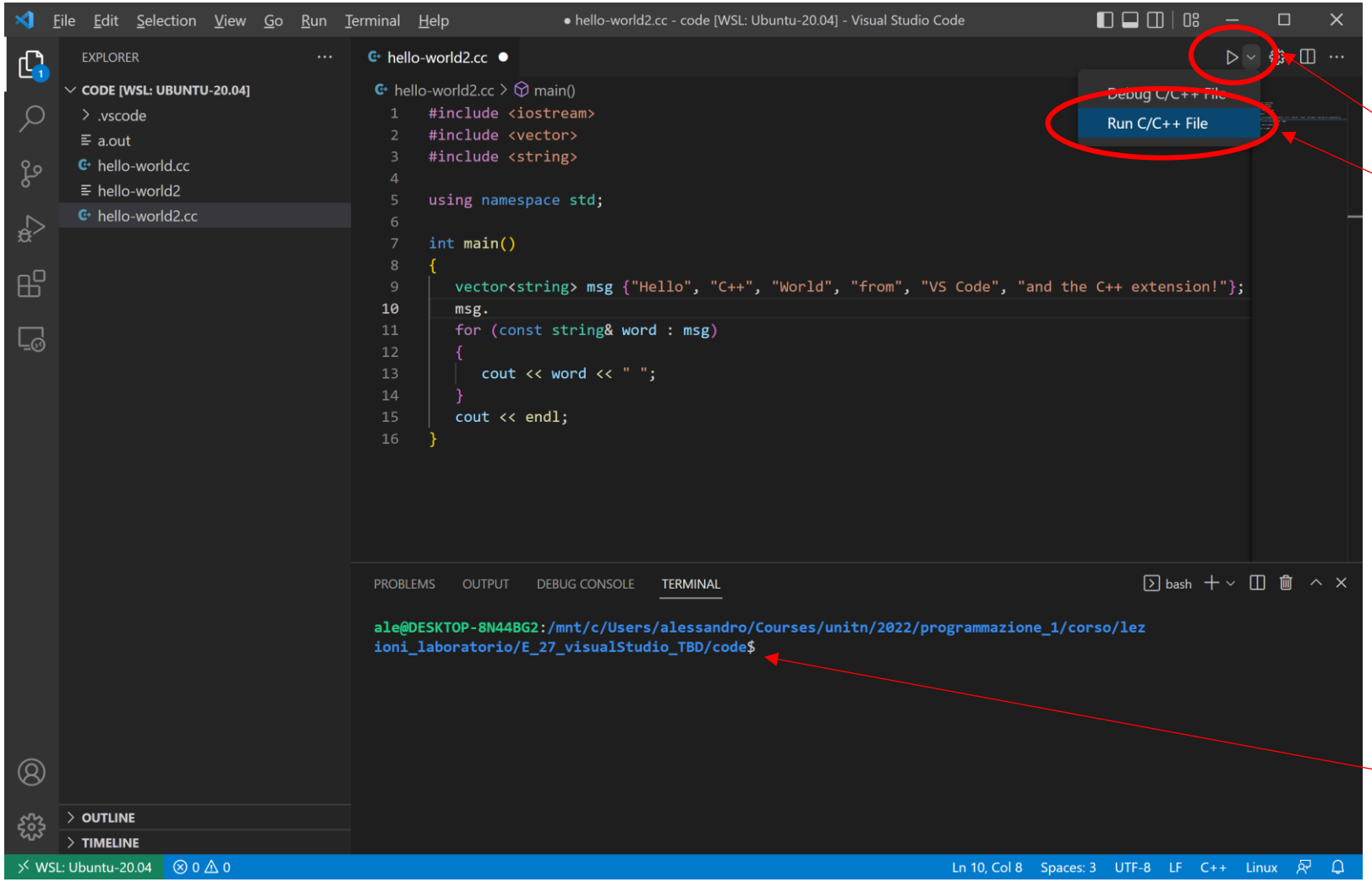
3.c) Write the code (4)

IntelliSense



Coding …
Use IntelliSense function for the automatic auto-completion of commands (shows the instruction availability)

3.d) Compile and Run the code (1)

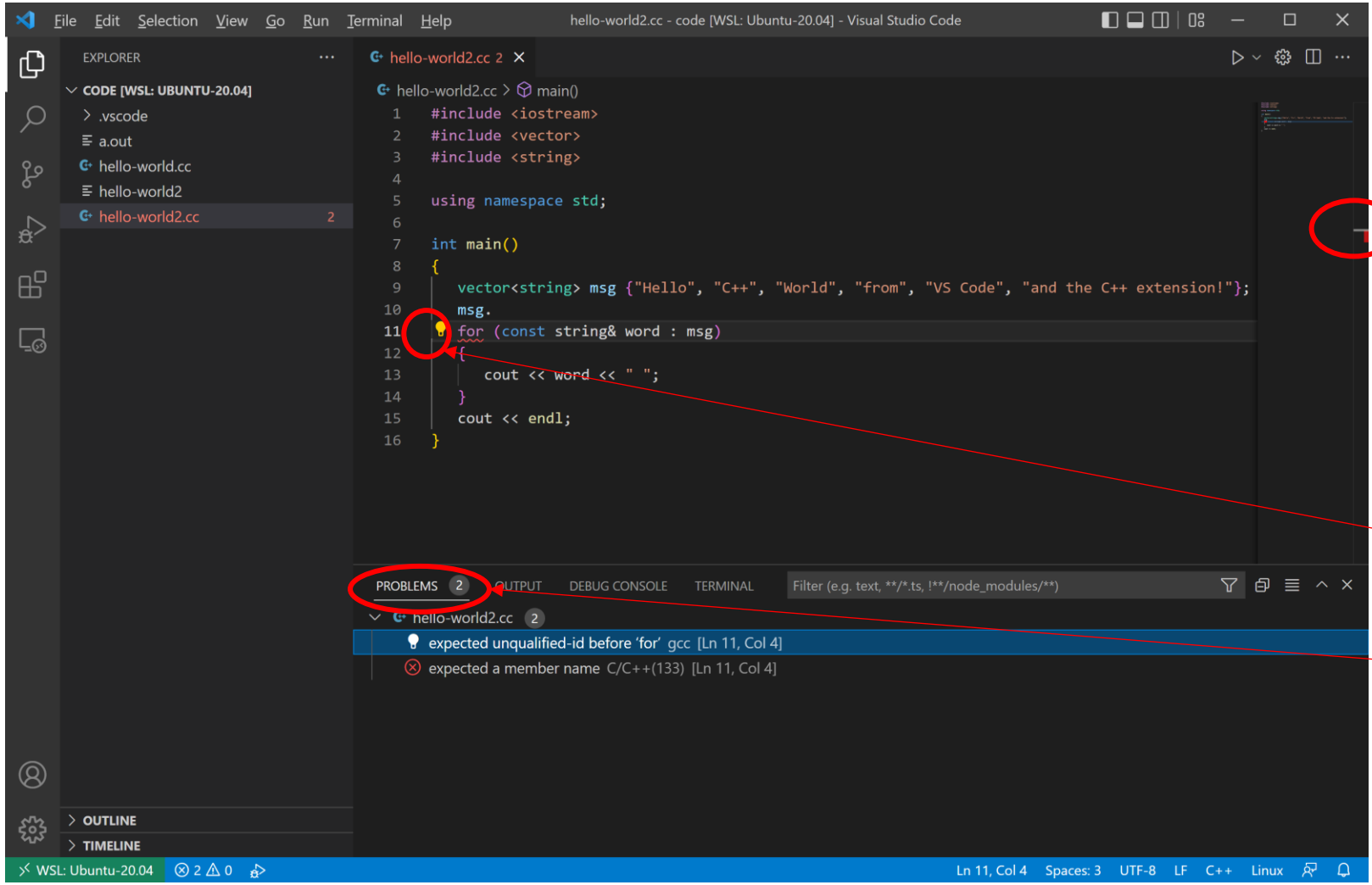


Start the compilation and execution

Compilation command via graphical interface

Compilation command in the Terminal
(1) go in the right folder, where the code is saved
(2) type the compile command «g++ hello-world2.cc»

3.e) Compile and Run the code (2)



Compilation error!

Compilation error
Details in tab Problems
Error line